SIVASHANKAR PALRAJ

Embedded Systems | Ultra-Low-Power IoT | Autonomy & Sensor Fusion

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CAREER SUMMARY

- Embedded systems engineer with **10+ years of experience** delivering **ultra-low-power IoT products**, Embedded C firmware, and production-ready PCB prototypes.
- Apply **probabilistic modelling, sensor fusion, and kinodynamic motion planning** to build risk-aware robotic systems for complex, dynamic environments.
- Developed practical autonomy capabilities, including **multi-sensor calibration and transform extraction**, for autonomous trucking applications.
- Design wired and wireless products with **EMC and SAR compliance** in mind, and improve delivery reliability through **CI/CD and RPA automation**.

EXPERIENCE

Lead Electronic Design Engineer

Electronic Temperature Instruments Ltd

Sep 2020 – Present Worthing, UK

- Lead the design and optimisation of ultra-low-power IoT devices, integrating embedded products with cloud APIs.
- Develop RTOS-based firmware for Wi-Fi data loggers and Bluetooth Low Energy (BLE) IoT devices.

Senior Autonomy Engineer

AiDrivers LTD

Jul 2017 – Jul 2020 Uxbridge, London.

- Developed ROS drivers for sensors, control drives, and actuators, supporting non-destructive testing (NDT) pose estimation.

Research Engineer

Electronics & Telecommunications Research Institute– MetaRPAS Inc.,

July 2016 – June 2017 Seoul, South Korea

- Contributed to agricultural UAS, warehouse management systems, AWS-connected solutions, 3D hologram projection with UAS, and motor-control algorithms.

Electronics Engineer

SyklarkDrones Pvt., Ltd.,

June 2014 – June 2016 Bangalore, India

- Delivered aerial data acquisition and analytics workflows, including point-cloud reconstruction.

EDUCATION

Master of Engineering in VLSI Design

Anna University

September 2012 – June 2014 Chennai, India

Bachelor of Engineering in Electrical Engineering

Anna University

September 2008 – May 2012 Chennai, India

PROFESSIONAL APPROACH

" Careful planning is as important as choosing the goal. "

TECHNICAL SKILLS

ROS | C++ | JS | RTOS | Low Power | Embedded

Git | CI/CD | DevOps | RPA | Autopilot | UAV/UAS

Sensor Fusion | RTK & VRS GPS

C *optimized* | VHDL | FPGA | MSP430 | ARM | Serial-Interfaces | WiFi | BLE5.0 | CAN – J1939

Matlab | Computer Vision | Python | Django

Project management | Requirement Analysis | Technical Write-ups | Digital Documentation | Safety Standards

LANGUAGES

Tamil
English
Telugu
Hindi
Korean

INTEREST

UAVs

Mathematics

Yoga